NMAP Basics

Nmap is an open-source utility for network exploration and security auditing.

Widely used to discover open ports, active hosts, running services & their versions.

Core functions :- Host discovery -> Identify which devices are online on the network

Port scanning -> Determine status of ports open/closed/filtered, scans like SYN scan (-sS) TCP connect (-sT), UDP scan (-sU).

Service & version detection :- (-sV) detects version of various services running on different ports.

Os detection :- (-O) to determine the Os of the target, based on TCP/IP fingerprinting

Basic Commands :- ① nmap <target ip> :- scans 1000 ports of the target machine.

② nmap -sV -sS -O <target ip> :- Conduct SYN scan, detect service versions & Os.

③ nmap -p <port no.> <ip> :- to scan specific port, on target

④ nmap -p- <target ip> :- scan all 65535 TCP ports on the target.

⑤ nmap -F :- scan top 100 common ports

Tips :- · run nmap scans with root privilages for advanced features like raw packet packet scanning.

· use -v to use verbose and see scan delails & progress.

· nmap -h to use help and guide for syntax.

Sample scan :-

```
└─$ sudo nmap -p 22 80 443 -v -O -sV <target ip>
[sudo] password for computer:
```